

Lukas Fleury

lfleury@uwaterloo.ca | (647) 923-5077 | Cambridge, Ontario | [LinkedIn](#) | [lukasfleury.com](#)

SKILLS

- **Software Tools:** SolidWorks (CSWA), AutoCAD, Inventor, Mastercam, MATLAB, C++, SAP, MS Office
- **Core Competencies:** CAD interpretation, DFM comprehension, Technical Drawing & GD&T

WORK EXPERIENCE

LiftWerx | Cambridge, Ontario | Operations Engineering Co-op 08/2026 - Present

- **Contribute** to the **design and implementation** of **up-tower lifting equipment** and **rigging systems**, enabling major turbine components to be serviced without full crane teardown.
- **Support** the design and development of **sustainable repair and maintenance solutions for wind turbines**, extending the service life and reliability of renewable-energy infrastructure.
- Assist in **planning and execution of wind turbine repair operations**, applying mechanical engineering principles to **balance safety, efficiency, logistics, and sustainability** across active wind farms.

Mammoet | Puslinch, Ontario | Engineering Co-op 01/2026 - 05/2026

- **Produced and revised 80+** fabrication drawings for **crane and heavy-lift equipment** using AutoCAD and AutoCRANE, improving drawing accuracy, standardization, and **database integration**.
- **Designed** detailed usage, assembly, and setup drawings for heavy-lift transport systems (jack-and-slide, SPMT interfaces), reducing field complexity and supporting safer, faster on-site execution.
- **Developed** project drawings to install a **150-ton** transformer at Bruce Power, and **100-ton** steam generators at Pickering Nuclear.

TECHNICAL PROJECTS

250 mm Custom-Built Radio-Controlled Arduino Quadcopter

- **Designed** and **built** a 250 mm quadcopter from scratch, integrating the **mechanical frame, electrical systems**, soldered connections, and **Arduino-based flight controller** within a fixed cost constraint.
- Conducted **quantitative flight** and **propulsion testing**, comparing 2- vs. 3-blade and low- vs. high-pitch propellers using thrust, hover endurance, step-respons to **optimize** airframe performance and controls.

Valvetronic-Style Dual-Exit Exhaust Conversion with Remote-Actuated Flow Control

- **Engineered a dual-exit exhaust assembly** with a custom **valvetronic-inspired flow-diverter** modeled in SolidWorks, integrating an **RF-controlled actuator** to toggle an exhaust valve for my personal vehicle.

OYAP Student Manufacturing Projects

- **Developed, analyzed** and **incorporated** challenging multi-part manufacturing projects to the **OYAP** manufacturing technology curriculum.
- **Created** structured, industry-style designs that improved technical skills and gave **60+** students experience in interpreting advanced blueprints while performing precise machining operations.

AWARDS

- 1st Place – Waterloo Regional Precision Machining Competition | SkillsOntario
- Engineering Certificate Program (ECP) | St. Benedicts C.S.S.

EDUCATION

University of Waterloo

Bachelor of Applied Science, Mechanical Engineering

Expected April 2030
